Training Day 16

1. Topics Covered and Brief Explanation

1. Introduction to Model Evaluation

Model Evaluation is the process of checking how accurately a machine learning model performs on unseen data. It helps determine whether a model is making correct predictions and identifies areas where its performance can be improved.

2. Accuracy

Accuracy measures the percentage of total predictions that are correct.

Accuracy = Correct Predictions / Total Predictions

3. Precision

Precision measures how many of the positive predictions made by a model are actually correct.

For example, in spam detection, it measures how many emails classified as spam are actually spam.

4. Recall

Recall measures how many actual positive cases are correctly identified by the model.

For example, in disease detection, it measures how many actual positive cases are successfully detected.

5. F1-Score

F1-score is the harmonic mean of precision and recall. It provides a combined measure of both metrics and is useful when the dataset has imbalanced classes.

6. Confusion Matrix

A confusion matrix is a table used to compare actual and predicted classifications. It contains four main values:

- True Positive (TP): Correctly predicted positive cases.
- True Negative (TN): Correctly predicted negative cases.
- False Positive (FP): Negative cases incorrectly predicted as positive.
- False Negative (FN): Positive cases incorrectly predicted as negative.

2. Tools and Technologies Used

Tool	Purpose
Python	Performing model evaluation.
Scikit-learn	Calculating evaluation metrics.

Pandas	Organizing actual and predicted values.
Matplotlib	Visualizing evaluation results.
Google Colab	Executing Python programs.

3. Practical Learning / Code Example

Example: Evaluating a Classification Model

```
from sklearn.metrics import (
    accuracy_score,
    precision_score,
    recall_score,
    f1_score,
    confusion_matrix)
y_actual = [1, 0, 1, 1, 0, 1]
y_pred = [1, 0, 1, 0, 0, 1]
print("Accuracy:", accuracy_score(y_actual, y_pred))
print("Precision:", precision_score(y_actual, y_pred))
print("Recall:", recall_score(y_actual, y_pred))
print("F1 Score:", f1_score(y_actual, y_pred))
print("Confusion Matrix:")
print(confusion_matrix(y_actual, y_pred))
```

Explanation:

- accuracy_score() calculates overall prediction accuracy.
- precision_score() calculates precision.
- recall_score() calculates recall.
- f1_score() calculates the F1-score.
- confusion_matrix() displays correct and incorrect predictions.

4. Learning Outcomes

- Understood the importance of evaluating machine learning models.
- Learned about accuracy, precision, recall and F1-score.
- Gained knowledge of the four components of a confusion matrix.
- Practised calculating evaluation metrics using Scikit-learn.